

XBee-Controlled Aircraft

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This is a remote-controlled aircraft project based on Arduino and XBee modules. The aircraft's movement is controlled through a brushless DC motor and three servo motors. It has an XBee-based remote control at the transmitter end and an Arduino-based aircraft at the receiver end. The aim of this project is to develop a 4-channel wireless remote control (RC) system with one XBee each in transmitter and receiver.

Circuit and working

Fig. 1 shows the circuit of XBee-controlled aircraft transmitter section and Fig. 2 shows its receiver section. The circuits are built around Arduino UNO (board1), a pair of XBee module (XB1

and XB2), electronic speed controller (ESC) module, three servo motors (M1-M3) and a BLDC motor (M4).

Transmitter section is driven by a 3.7V battery and receiver section by

an 11.1V LiPo battery. 11.1V is used to power the brushless DC motor (or BLDC motor) through ESC module. The Arduino board1 receives power supply from 11.1V battery at its Vin input pin.

The four potentiometers in transmitter section are used for sending different control signals to the receiver section through RF modules. Control signals received in the receiver are processed by micro-controller in the Arduino, which

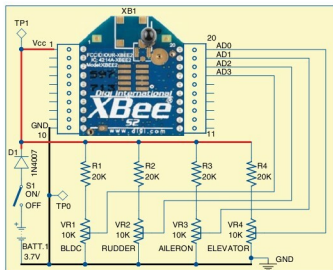


Fig. 1: Circuit of XBee-controlled aircraft (transmitter side)

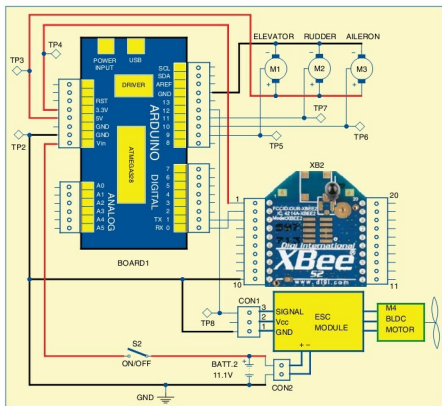


Fig. 2: Circuit of XBee-controlled aircraft (receiver side)

Test Points

Test point	Details
TP0	0V
TP1	3.1V (with respect to TP0)
TP2	0V
TP3	5V (with respect to TP2)
TP4	3.3V (with respect to TP2)
TP5	PWM signals for servo motor M1
TP6	PWM signals for servo motor M3
TP7	PWM signals for servo motor M2
TP8	PWM signals for BLDC motor M4

PARTS LIST

Semiconductors:

- Board1 - Arduino UNO board
- XB1, XB2 - XBee module
- ESC module - ESC module

Resistors (all 1/4-watt, $\pm 5\%$ carbon):

- R1-R4 - 20-kilo-ohm
- VR1-VR4 - 10-kilo-ohm preset

Miscellaneous:

- CON1 - 3-pin connector
- CON2 - 2-pin terminal block connector
- BATT.1 - 3.7V battery
- BATT.2 - 11.1V LiPo battery
- M1-M3 - 5V servo motor
- M4 - BLDC motor
- S1, S2 - On/off switch
- Propeller blades - Aircraft parts